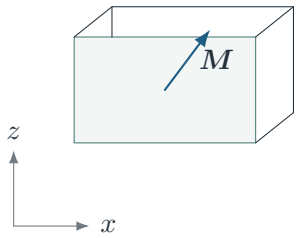
16-W1　均匀磁化长方体的表面电流

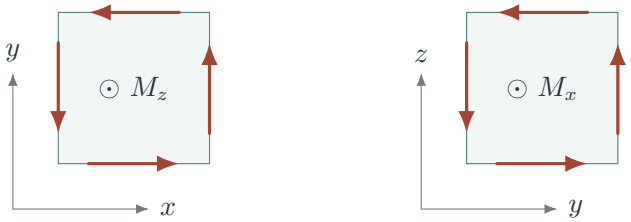


M_x, M_z 为常量

六个面分别取外法向

图 16-9　题图　长方体各面按同一组坐标轴定义方向

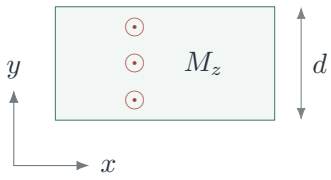
绕 z 轴的表面环流　　绕 x 轴的表面环流



正分量示意；负分量时相应环流反向，$\boldsymbol{K}_b = \boldsymbol{M} \times \boldsymbol{n}$

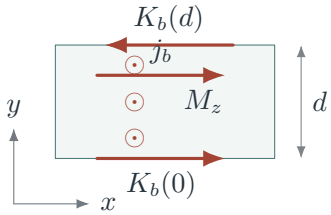
图 16-10　解答分析图　分别叠加绕 x 轴与 z 轴的两组闭合表面环流

16-W2　磁化沿厚度变化的平板



$$M_z = M_0(1 + y/d)$$
$$0 < y < d$$

图 16-11　题图　磁化朝纸外，其大小随厚度增大



$$j_x = M_0/d$$
$$K_x(0) = M_0,\ K_x(d) = -2M_0$$
$$B_z = \mu_0 M_z,\ H = 0$$

图 16-12　解答分析图　表面电流和体电流合计为零，板外附加场相消

16-W3　有限磁化圆柱的中心场

均匀给定磁化 M

中心 $z = 0$

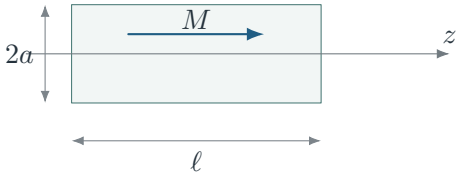


图 16-13　题图　圆柱轴向取为 z，侧面环流按右手方向

$$\mathrm{d}I_b = M\,\mathrm{d}z'$$

$$\mathrm{d}B_z(0) = \frac{\mu_0 M a^2\,\mathrm{d}z'}{2(a^2+z'^2)^{3/2}}$$

$$H_b(0) = B_b(0)/\mu_0 - M$$

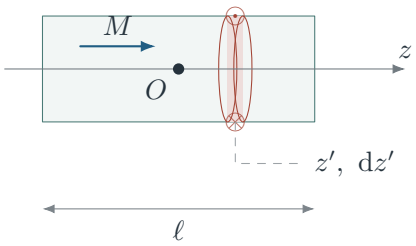
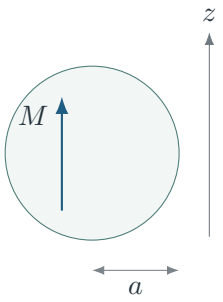


图 16-14　解答分析图　上侧朝纸外、下侧朝纸内的面电流产生向右的附加场

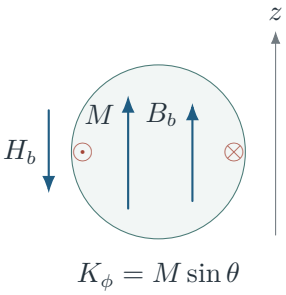
16-W4　均匀磁化球的内外场



给定磁化，非诱导磁化

球外真空

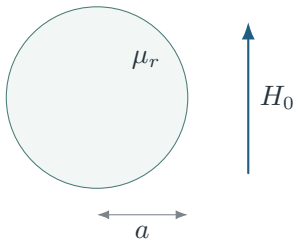
图 16-15　题图　球内磁化向上，球外场在无穷远消失



$$H_b = -M/3$$
$$B_b = 2\mu_0 M/3$$
$$m = VM$$

图 16-16　解答分析图　截面电流绕轴闭合，B 与 H 在球内方向相反

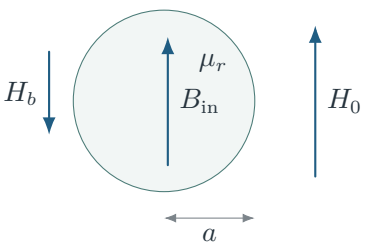
16-W5　线性球的磁化与表观磁化率



$$M = \chi_m H$$

真空外场 $B_0 = \mu_0 H_0$

图 16-17　题图　各向同性线性球在均匀外场中磁化



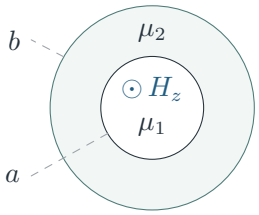
$$H = H_0 - M/3$$

$$M = 3(\mu_r - 1)H_0/(\mu_r + 2)$$

$$B/B_0 = 3\mu_r/(\mu_r + 2)$$

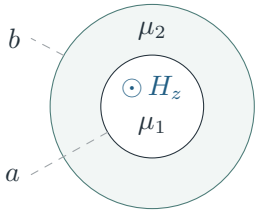
图 16-18　解答分析图　图示顺磁情形，退磁场与磁化反向

16-W6　径向分层的长螺线管



无限长螺线管横截面
自由绕组在 $r = b$

图 16-19　题图　内柱与环层沿半径分区，共用同一自由绕组



无限长螺线管横截面
自由绕组在 $r = b$

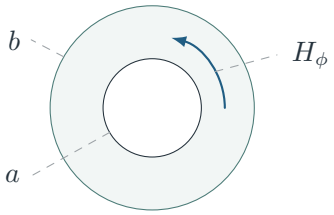
$H_z = nI$（内部两区）

$K_\phi(a) = M_1 - M_2$

$K_\phi(b) = M_2$

图 16-20　解答分析图　H 的轴向分量连续，两个分区的磁通相加

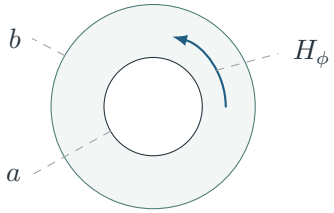
16-W7　环形线圈的有限径向厚度



N 匝，电流 I

高度 h，磁导率 μ

图 16-21　题图　环形线圈的内外半径均进入精确积分



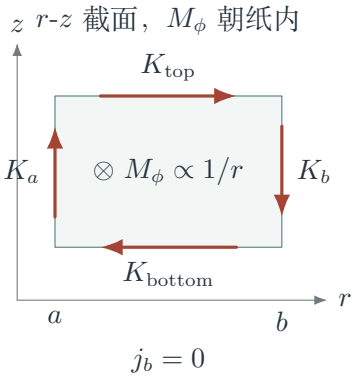
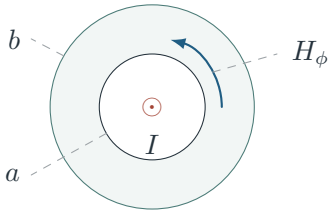


图 16-22　解答分析图　并列环形截面与 r-z 截面，明确体电流及内外、上下四面的闭合方向

16-W8　渐变磁化率圆柱壳

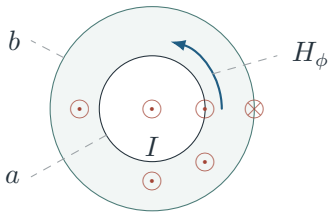


横截面，$+z$ 朝纸外

壳内 $\chi_m = \chi_0 r^2/a^2$

中心导线：自由电流 I

图 16-23　题图　渐变磁化率壳层围绕轴线自由电流



$\odot: +z, \quad \otimes: -z$

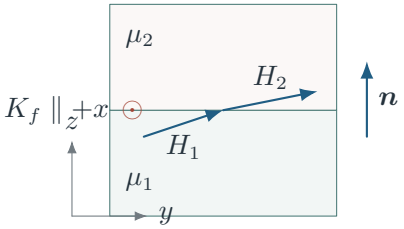
$j_{b,z} = \chi_0 I/(\pi a^2)$

内面正向，外面反向

壳外束缚电流净贡献为零

图 16-24　解答分析图　图按正磁化率绘制体电流和两面电流的方向

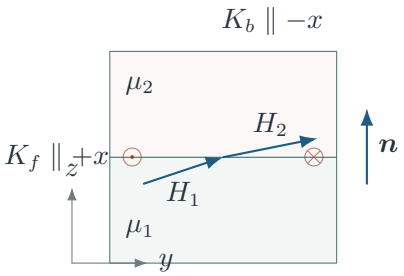
16-W9　带自由面电流的磁介质界面



$$K_f = H_* e_x$$

$$\mu_1 = 2\mu_0,\ \mu_2 = 5\mu_0$$

图 16-25　题图　在 yz 截面中，自由面电流朝纸外的正 x 方向



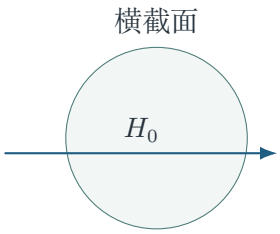
$$H_{2y} = 2H_*,\ H_{2z} = 2H_*/5$$

$$K_b = -5H_* e_x$$

$$K_{\text{tot}} = -4H_* e_x$$

图 16-26　解答分析图　净束缚电流朝负 x，H 切向跳变只计自由项

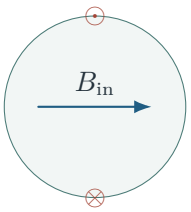
16-W10　横向外场中的无限长圆柱



外场垂直于柱轴

$$\mu = \mu_0\mu_r$$

图 16-27　题图　z 轴朝纸外，外场沿 x 方向



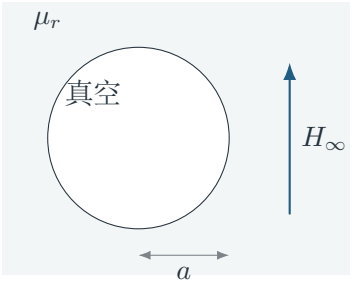
$$H_{\text{in}} = 2H_0/(\mu_r + 1)$$

$$H_b = -M/2$$

$$K_z = M\sin\phi$$

图 16-28　解答分析图　图示顺磁情形，上半周电流朝纸外

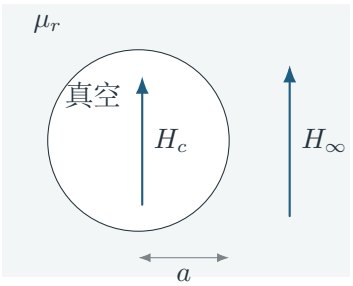
16-W11　磁介质中的球形真空腔



材料磁化重新平衡

远处 $B_\infty = \mu H_\infty$

图 16-29　题图　白色为球形真空腔，浅色仅示意向远处延伸的磁介质

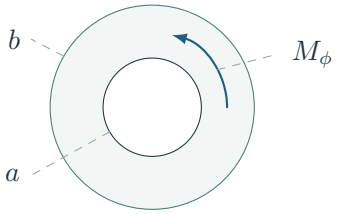


$$H_c/H_\infty = 3\mu_r/(1+2\mu_r)$$

$$B_c/B_\infty = 3/(1+2\mu_r)$$

图 16-30　解答分析图　腔内 H 与 B 分别和远处对应量比较

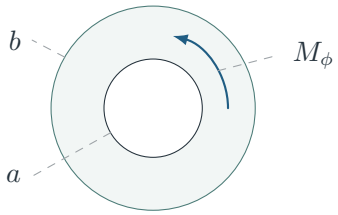
16-W12　剩磁环的体电流与表面电流

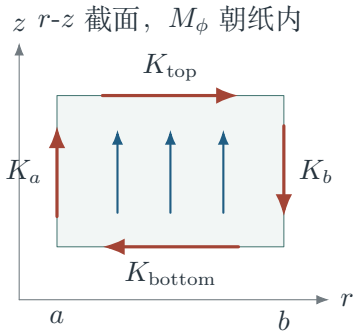


$M_\phi = M_0$ 为常数

无自由电流

图 16-31　题图　周向磁化取逆时针，环内没有外加绕组

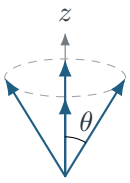




$j_{b,z} = M_0/r$

图 16-32　解答分析图　并列环形截面与 r-z 截面，明确体电流及内外、上下四面的闭合方向

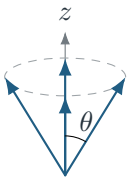
16-T1　磁矩取向与宏观磁化



各磁矩大小 m_a

方位角均匀分布

图 16-33　题图　圆锥取向只留下沿轴的平均磁矩



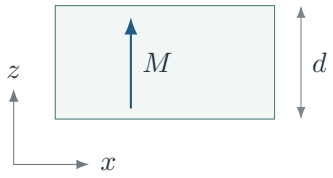
横向平均为零

$M_z = nm_a\cos\theta$

两种取向：$M_z = nm_a(2p-1)$

图 16-51　解答分析图　磁矩取向分布决定磁化，个体磁矩不因平均为零而消失

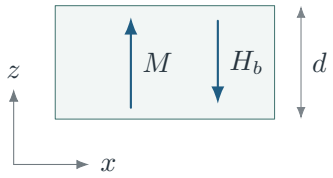
16-T2　法向磁化的无限平板



给定法向磁化 M_0

板外无外加场

图 16-34　题图　磁化垂直于无限平板



$j_b = K_b = 0$

$B = 0$（全部空间）

板内 $H = -M_0 e_z$

图 16-52　解答分析图　B 为零，板内 H 仍由磁化定义产生非零退磁场

16-T3　转向磁化平板

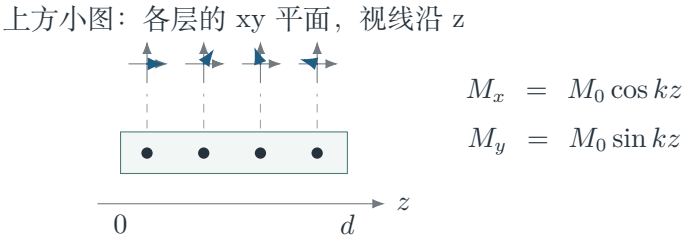


图 16-35　题图　厚度方向的磁化取向连续转动，箭头为各层方向示意

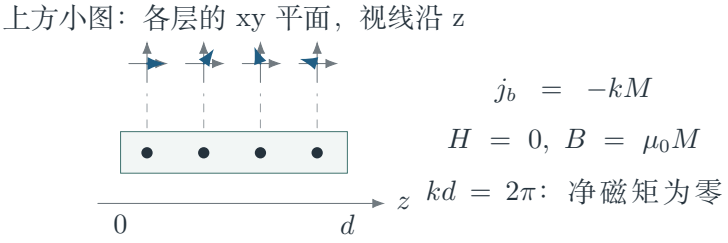


图 16-53　解答分析图　磁化的旋度非零，即使转一周后的净磁矩为零

16-T4　有限圆柱的远场

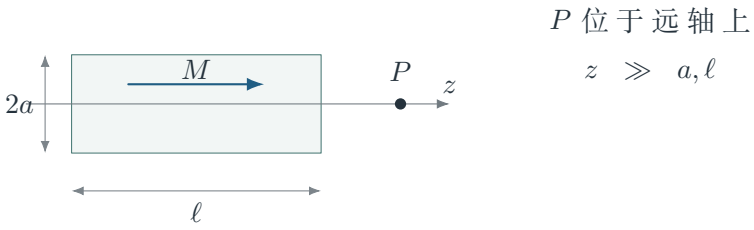


图 16-36　题图　远处 P 点只取偶极近似的首项

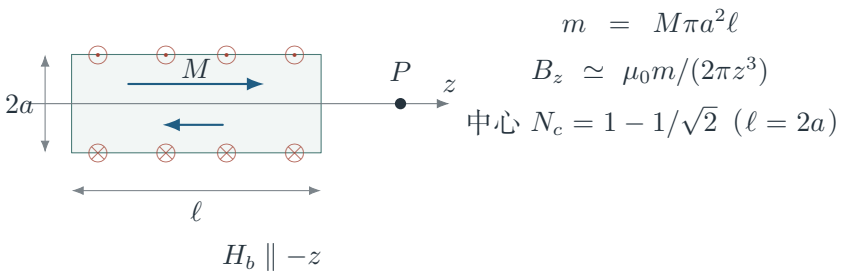
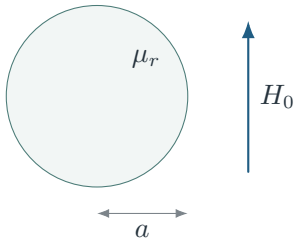


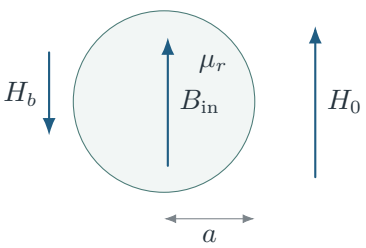
图 16-54　解答分析图　用环流积分与总磁矩两种方法核对远场

16-T5　由球内磁感应反推材料参数



测得 $B_{\text{in}}/B_0 = 3/2$

图 16-37　题图　已知的是球内 B 相对于真空外场的比值



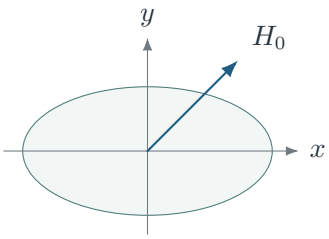
$\mu_r = 2,\ \chi_m = 1$

$H/H_0 = M/H_0 = 3/4$

表观磁化率低估 25%

图 16-55　解答分析图　先消除形状退磁影响，再反推材料磁化率

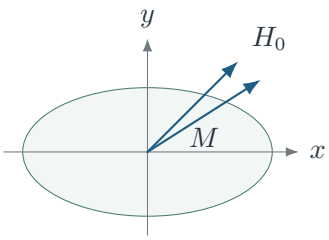
16-T6　斜置椭球的磁化



外场方向 45°

材料各向同性，$\chi_m = 4$

图 16-38　题图　主轴方向有不同的形状退磁因子



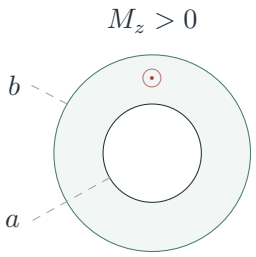
$N_x = 1/10,\ N_y = 3/10$

$M_y/M_x = 7/11$

磁化偏向退磁较弱的 x 轴

图 16-56　解答分析图　内部 M 相对外场发生偏转，局部 M 与 H 仍平行

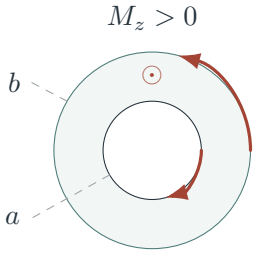
16-T7　轴向磁化的空心长圆柱



轴向磁化，磁化冻结

无限长圆柱壳

图 16-39　题图　磁化朝纸外，壳内与壳外都是真空



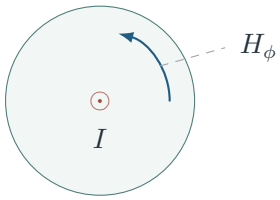
内面 $K_\phi = -M$

外面 $K_\phi = +M$

腔内 $B_b = 0$

图 16-57　解答分析图　内外电流柱面的内腔场等量反向

16-T8　有均匀自由体电流的磁性导线

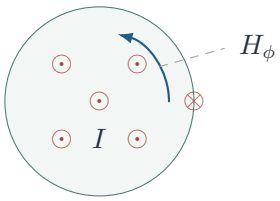


自由电流在面积内均匀

磁性导线 μ_r

横截面，$+z$ 朝纸外

图 16-40　题图　整个磁性圆柱同时承载自由体电流



$j_b = \chi_m j_f$

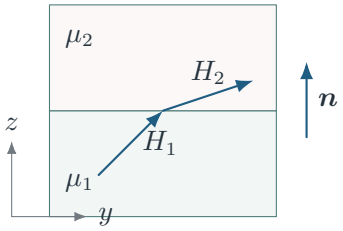
体束缚电流 $\chi_m I$

表面束缚电流 $-\chi_m I$

$\odot: +z,\quad \otimes: -z$

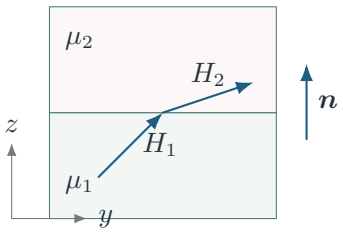
图 16-58　解答分析图　图示正磁化率，体电流与表面回流相消

16-T9　磁场线跨越无电流界面



$$\mu_2/\mu_1 = 3$$
$$\theta_1 = 45^\circ$$

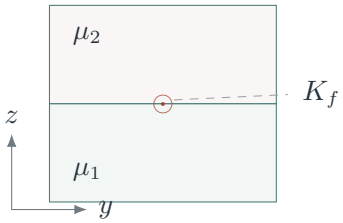
图 16-41　题图　无自由面电流的界面连续 Ht 与 Bn



$$H_{2y} = H_{1y}$$
$$H_{2z} = H_{1z}/3$$
$$\tan\theta_2 = 3$$

图 16-59　解答分析图　上侧的切向与法向分量之比为 3，场线更偏向界面

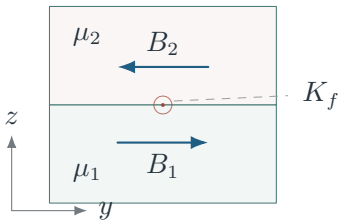
16-T10　自由电流片与两个半空间



无附加均匀场

yz 截面，$+x$ 朝纸外

图 16-42　题图　自由面电流位于两个线性半空间之间



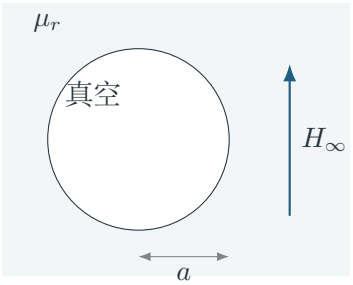
$$B_1 = -B_2$$

$$H_{1y} - H_{2y} = K$$

$$|B| = \mu_1\mu_2 K/(\mu_1 + \mu_2)$$

图 16-60　解答分析图　磁感应等量反向，H 的大小随两侧磁导率而不同

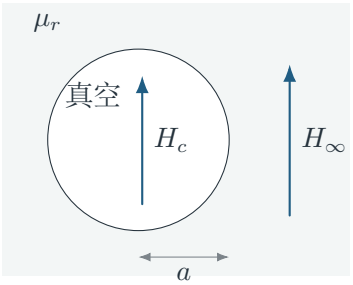
16-T11　球腔中两个场的比值



$$\mu_r = 5$$

给定远处 B_∞

图 16-43　题图　白色为球形真空腔，浅色仅示意向远处延伸的磁介质



$$H_c/H_\infty = 15/11$$

$$B_c/B_\infty = 3/11$$

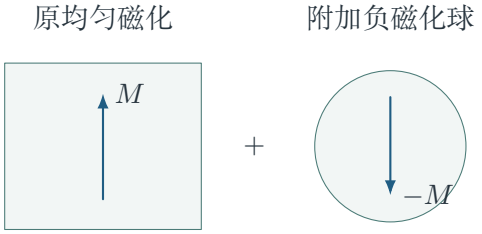
图 16-61　解答分析图　球腔不含磁化，腔内 B 由真空磁导率连接 H

16-T12　冻结磁化中的球形孔洞



孔外磁化冻结
自由电流保持不变

图 16-44　题图　从冻结磁化背景中移去一个均匀磁化球



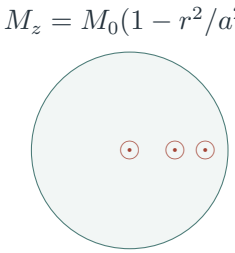
$$\Delta B = -2\mu_0 M/3$$

$$B_c = B_{\text{bulk}} + \Delta B$$

球外磁化冻结；自由电流保持不变

图 16-62　解答分析图　负磁化球的附加场与原场相加

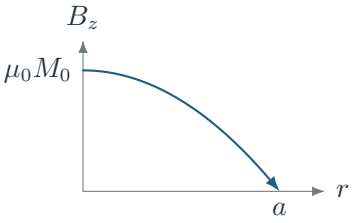
16-T13　没有侧面电流的磁化圆柱



磁化沿轴，边缘降为零

无限长圆柱

图 16-45　题图　圆柱内的磁化朝纸外，大小沿半径减小



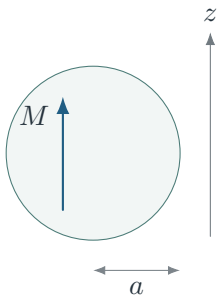
$j_\phi = 2M_0 r/a^2$

$K_b(a) = 0$

内部体环流产生轴向场

图 16-63　解答分析图　轴向磁感应沿半径呈抛物线下降，表面连续至零

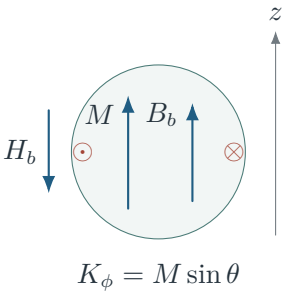
16-T14　均匀磁化球的两种能量



给定永久磁化

分别积分 H_b^2 与 B_b^2

图 16-46　题图　球内 B 与 H 的定义不同，能量积分须明确所用场



$K_\phi = M\sin\theta$

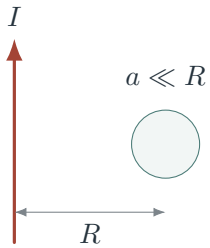
$W_d = \mu_0 V M^2/6$

$W_B = \mu_0 V M^2/3$

球外两种积分相同

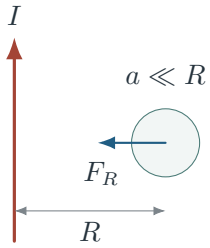
图 16-64　解答分析图　退磁能和真空形式 B 场能都需包括球外偶极场

16-T15　直导线附近的小磁球



导线电流固定
球的磁矩由外场诱导

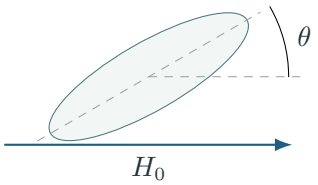
图 16-47　题图　球心远离导线，小球可采用局部均匀外场



$m = \alpha I/(2\pi R)$
图示 $\alpha > 0$：吸引
$\alpha < 0$ 时反向

图 16-65　解答分析图　力指向高 H^2 区域或低 H^2 区域，取决于有效极化率符号

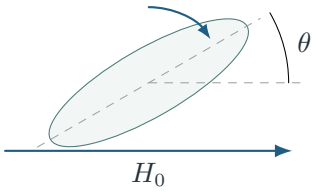
16-T16　细长磁性椭球的取向



长轴退磁因子较小

本征磁化率各向同性

图 16-48　题图　θ 从外场方向量向椭球长轴



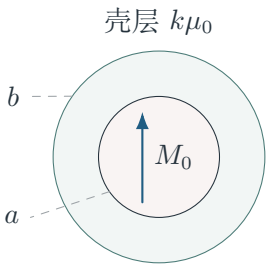
$$\alpha_{\parallel} > \alpha_{\perp}$$

$$\tau = -\mu_0 H_0^2 \Delta\alpha \sin 2\theta/2$$

顺磁与抗磁均趋向长轴平行

图 16-66　解答分析图　小正角度的力矩使 θ 减小

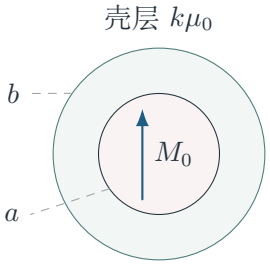
16-T17　永久磁体的磁性包覆层



芯磁化固定，壳磁化诱导

没有自由电流

图 16-49　题图　两种不同磁化约束分别进入内界面方程



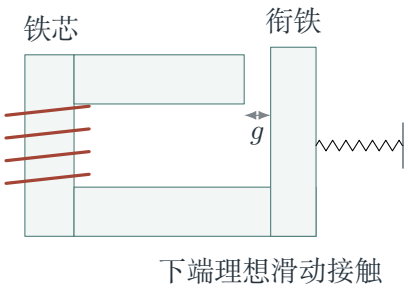
芯：$B = \mu_0(H + M_0)$

壳：$B = k\mu_0 H$

外部：偶极矩 $4\pi E$

图 16-67　解答分析图　先解四个边界方程，再求壳层磁化和面电流

16-T18　含气隙铁芯的吸合失稳

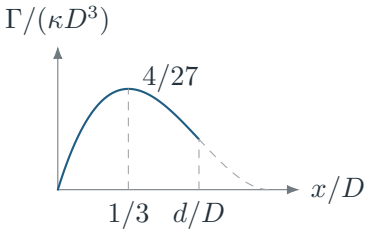


单气隙 $g = d - x$

铁芯固定，衔铁向左位移 x

忽略滑动接触的额外磁阻

图 16-50　题图　单气隙缩短 x，下端滑动接触在本模型中忽略额外磁阻



峰值左侧：稳定

峰值右侧：不稳定

固定磁链时力为常量

图 16-68　解答分析图　图取 d/D=2/3 示意；接触点后的虚线是方程的数学延拓